Physics

Ans- 1.- Convex lens

Ans-2 Convex lens of 10cm focal length has greater power

Ans-3 focal length is define as the distance between the centre of the lens or mirror and its focal ~~length~~ point

$u = -5\,cm$

$v = 10\,cm$

Using lens formula

$$\frac{1}{f} = \cancel{\frac{1}{u}}\ \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{(-5)}$$

$$\frac{1}{f} = \frac{3}{10} \qquad \boxed{f = \frac{10}{3}}$$

~~f0~~

Ans-4 Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Here v and u are the position of the image and object with respect to lens.

Magnification is the ratio of size of the image and the size of the object

$$m = \frac{v}{u}$$

Hence image is inverted

$$-1 = \frac{v}{u}$$

$$v = -u$$

Given that image is formed at the distance from +50 from the lens. And since the image is real

$v = +50$ and $u = -50$

$$\frac{1}{f} = \frac{1}{50} - \frac{(-1)}{50}$$

$$\frac{1}{f} = \frac{2}{50} = \frac{1}{25}$$

$f = 25$.

Power of the $P = \frac{1}{f}$

$$P = \frac{1}{25} \times 10^{-2} m^{-1}$$

$$P = \frac{100}{25} = 4 m^{-1}$$

~~P =~~ $1 m^{-1} = 1D$

So ~~P = 4D~~ $\boxed{P = 4D}$